| | 1 A | 2 B | 3 C | 4 D | 5 E |
|---|---|---|---|---|---|
| 1 ABC | $a_1$ | $a_2$ | $a_3$ | $b_{14}$ | $\cancel{b_{15}}$ $a_5$ |
| 2 AD | $a_1$ | $\cancel{b_{22}}$ $a_2$ | $b_{23}$ | $\cancel{a_4}$ $b_{14}$ | $b_{25}$ |
| 3 BDE | $b_{31}$ | $a_2$ | $b_{33}$ | $a_4$ | $a_5$ |

✓ $A \to BD$

$a_1 \to a_2 \quad b_{14}$

$a_1 \to \cancel{b_{22}}\ a_2 \quad \cancel{a_4}\ b_{14}$

✓ $B \to E$

$a_2 \to \cancel{b_{15}}\ a_5$

$a_2 \to a_5$

Lossy Decomp

$A \to BD$

$A \to B$

$A \to D$

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$A \to B,\ A \to D,\ B \to E$

| | A | B | C | D | E |
|---|---|---|---|---|---|
| ABC | $a_1$ | $a_2$ | $a_3$ | $\cancel{b_{14}}$ $a_4$ | $\cancel{b_{15}}$ $a_5$ |
| AD | $a_1$ | $\cancel{b_{22}}$ $a_2$ | $b_{23}$ | $a_4$ | $\cancel{b_{25}}$ $a_5$ |
| BDE | $b_{31}$ | $a_2$ | $b_{33}$ | $a_4$ | $a_5$ |

All a's

Lossless Decomp

$A \to B$

$a_1 \to a_2$

$a_1 \to \cancel{b_{22}}\ a_2$

$A \to D$

$a_1 \to \cancel{b_{14}}\ a_4$

$a_1 \to a_4$

$B \to E$

$a_2 \to \cancel{b_{15}}$

$a_2 \to \cancel{b_{25}}$

$a_2 \to a_5$